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HENNAI-HEADQUARTERED

automobile major TVS Motor Company uses high-performance computing (HPC) for running R&D simulations and testing the aerodynamics of two-wheelers, which allows it to make the vehicles stable at speed and more efficient, cool engines faster, and achieve better top

speeds, quicker. It has been doing this for, like, forever, but something changed in 2020. Earlier, TVS would purchase HPC hardware upfront for five-year periods, to be installed, run and maintained by its own team. As Ranjith Radhakrishnan K.C., AVP-IT, TVS Motor Company, informs us, in this 'capex' model, "we had to actually pay upfront for five years, and when we revised something, we had to make sure that it lasted five years".

In 2020, TVS switched to HPE GreenLake's HPC-as-a-Service (HPCaaS), an 'opex' model perched on its premise. Ergo, pay per use, and all other attendant cloud benefits. "Since this HPCaaS model is scaleable, we don't have to invest in what we may require after two-three years, or even more," says Radhakrishnan. "We are able to invest only in whatever is required for this current financial year." Another perk is the reduction in lead time for ordering new hardware, which has come down to about four weeks from eight weeks earlier. "This model enabled us to be much more nimble as we don't have to plan two-three months in advance." Explains Sinisa Nikolic, Director & Segment

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The HPC Market

The HPCaaS segment of the HPC market is growing at a much faster speed than other segments

- ➔ HPCaaS market is \$4.5 billion globally and expected to double to \$9 billion by 2025
- ➔ HPCaaS in India is expected to grow to around \$102 million by 2025 from \$47 million in 2020
- ➔ In 2020, the overall HPC server market globally was \$13.5 billion. In 2025, this is expected to climb to \$20 billion
- ➔ Globally on-premise HPC market to reach \$40 billion with add-on storage, middleware, applications, and maintenance services

Source: Hyperion Research

The Pricing

HPC on the cloud comes at very attractive costs compared to an on-premise setup



Most Economical plan

₹11,000 per month
Compute power: 4 vCPU, 20 GB RAM, 8 GB NVIDIA V100 Tensor Core GPUs, 250 GB DISK, vNIC and up to 10G Network performance with Unlimited Data Transfer and Host Anti-Virus



Most Premium plan

₹1.74 lakh per month
Compute power: 2 x Intel Xeon Gold 6230 2.1GHz, 20C, 12x64 (768) GB RAM, 2x480 GB HDD, 4x10G & 2x1G NIC, 4 x NVIDIA V100 Tensor Core GPUs 32GB and full height GPU with 36 months Pro Support



On-prem setup

An on-premise setup with tech specs comparable to the "most economical plan" above could cost an organisation upwards of ₹50 lakh. There will also be additional cost factors like power, cooling, network, manpower and software licences

Source: Yotta